MIL-A-85097/3B(AS)
17 October 1985
SUPERSEDING
MIL-A-85097/3A(AS)
3 July 1980

MILITARY SPECIFICATION SHEET

AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS); ROCKET MOTOR (LOADED ASSEMBLY) MARK 79 MODS

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the Mark 79 Mods Rocket Motors described herein shall consist of this specification and the latest issue of MIL-A-85097.

1. SCOPE

1.1 Scope. This specification sheet covers requirements unique to the Rocket Motors Mark 79 Mod 0 and Mod 1 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following document is in addition to those listed in the basic specification, MIL-A-85097, and form a part of this specification sheet to the extent specified herein.

DRAWINGS

Naval Air Systems Command (Code Ident 30003)

672AS200	Rocket Motor, Mark 79 Mods

3. REQUIREMENTS

3.1 General. The requirements in this specification sheet, unless in direct conflict with the basic specification, shall be considered additional, not substitute. The Mark 79 Mods Rocket Motor shall conform to Drawing 672AS200.

3.2 Preproduction.

3.2.1 Propellant aging. When tested in accordance with MIL-A-85097, the propellant shall meet the requirements made available by the contractor and accepted by the Government.

3.3 Preproduction/production.

3.3.1 Propellant. The propellant shall conform to the contractor's in-house propellant specification when tested for the data specified in MIL-A-85097. The contractor shall make available the results of the tested values and the in-house specification limits for comparison.

3.3.2 Ballistic performance. The ballistic performance shall conform to table I.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1377

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:23Z
Check the source to verify that this is the current version before use.

TABLE I. Ballistic performance requirements.

Characteristic	Value Min	Value Max	Temperature (± 5°F)
Total impulse (lb-s)	590	640	−40, 160
Action time (ms)	190	320	−40, 160
Thrust (lb)	-	3800	160
Ignition delay (ms)	-	25	−40, 160
Initiation pressure (psig)	400	650	−40, 160
Time to shear (ms)	-	10	−40, 160

3.3.2.1 Rise time. The rise time shall be not greater than 30 milliseconds (ms) from 10 percent of maximum thrust to 75 percent of maximum thrust on the rising portion of the thrust-time curve.

3.3.2.2 Chamber pressure. On preproduction units only, combustion chamber pressure versus time shall be recorded.

4. QUALITY ASSURANCE PROVISIONS

4.1 Preproduction/production.

4.1.1 Conditioning temperatures. The conditioning temperatures and tolerance shall be as follows:

a. Cold: −40° ± 5°F
b. Hot: 160° ± 5°F

4.1.2 Conditioning period. Time for conditioning prior to static firing or during temperature cycling shall be as follows:

a. Static firing: Not less than 4 hours
b. Temperature cycling: Not less than 5 hours

4.1.3 Radiographic inspection. Radiographic inspection shall be as follows:

a. Inspect the propellant grain assembly at the head and aft end. Four longitudinal views shall be taken at each end, 45° apart. The grains shall be oriented so the 0° and 90° planes pass through the slots.

b. Inspect the rocket motor assembly in two views 90° apart.

5. PACKAGING

(See latest revision of MIL-A-85097.)

6. NOTES

6.1 Intended use. The rocket motor is intended for use with an ejection seat system as a means to propel the occupant of an aircraft safely out of and away from the aircraft while in the seat and at sufficient velocity to ensure that the occupant will clear a fast moving aircraft by a safe margin.

6.2 Changes from previous issue. Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - AS
(Project 1377-N769)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:23Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-85097/3B(AS)	Aircrew Escape Propulsion Systems (AEPS); Rocket Motor MARK 7

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

Source: https://assist.dla.mil – Downloaded: 2026-10-23T10:23Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY
Commanding Officer
Naval Air Engineering Center
Systems Engineering Standardization Department
(SESD), Code 93
Lakehurst, NJ 08733

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer
Naval Air Engineering Center
Systems Engineering Standardization Department
(SESD). Code 93
Lakehurst, NJ 08733

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:23Z
Check the source to verify that this is the current version before use.